

Assignment 2

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Task 1:

1. data_bus_32: Valid
2. \$display_enable: Invalid - not a normal identifier.
3. _cdc_sync_01: Valid
4. \cpu_reset_n: Valid escaped identifier.
5. SoC-AXI-Interface: Invalid because '-' is not allowed.
6. wire_internal-sig: Invalid because '-' is not allowed.
7. module: Invalid because it is a keyword.
8. Clock_50MHz: Valid
9. \display_val: Invalid in this usage.
10. 1st_data_line: Invalid because it starts with a number.
11. \reset#_n: Valid escaped identifier.
12. \$display_val: Invalid.
13. DATA and data are different variables because Verilog is case sensitive.

Task 2:

1. 16'hFF0A
2. 8'bxxxx_zzzz
3. 4096 and 32'd4096
4. 4'sb1011
5. -4096 and -32'd4096
6. 1.25e-4
7. 3.141592653589793
8. 4'o12

Task 3:

Literal	Width	Base	Binary	Decimal
8'hA3	8	Hex	10100011	163
6'h2x	6	Hex	10xxxx	Unknown
4'b101z	4	Binary	101z	Unknown
12'o47	12	Octal	000000100111	39
8'd255	8	Decimal	11111111	255
'h1F	Unsize	Hex	11111	31
-15	Unsize	Decimal	-15	-15
325	Unsize	Decimal	325	325
5'sd10	5	Signed Decimal	01010	10
3'sd12	3	Signed Decimal	100	-4
-6'd7	6	Decimal	111001	-7
10'bx0_101z	10	Binary	x0101z	Unknown
5'shC	5	Signed Hex	01100	12

Part 2:

```
module reset_sync(  
    input wire clk,  
    input wire async_rst_n,  
    output reg sync_rst_n  
);  
    reg meta_reg;  
    reg sync_reg;  
    parameter DELAY = 8'd15;  
    parameter MAX_VAL = 4'b1010;  
    always @(posedge clk or negedge async_rst_n) begin  
        if (!async_rst_n) begin  
            meta_reg <= 1'b0;  
            sync_reg <= 1'b0;  
        end  
        else begin  
            meta_reg <= 1'b1;  
            sync_reg <= meta_reg;  
        end  
    end  
    assign sync_rst_n = sync_reg;  
  
endmodule
```

Part 3:

```
/*
Lab: Verilog Lexical Conventions
Description: Demonstrates identifiers, numbers, strings, and attributes.
*/

(* optimize = "off" *)
module lexical_demo;

// 1. Strings
// "Hello World" contains 11 characters
reg [87:0] my_string;

// 2. Case Sensitivity & Identifiers
reg [7:0] Data_Val;
reg [7:0] data_val;

// 3. Logic Values & Literal Integers
reg [11:0] mixed_logic;

// 4. Literal Real Numbers
real analog_voltage;

// 5. Attributes on variables
(* keep = "true" *) wire protected_wire;

initial begin

// String assignment
my_string = "Hello World";

// Assign different values to case sensitive variables
Data_Val = 8'hAA;
data_val = 8'h55;

// Binary value containing 1,0,x,z
mixed_logic = 12'b1010_xxxx_zzzz;
```

```
// Scientific notation real value
analog_voltage = 3.3e-3;

// Display Results
$display("--- Lexical Conventions Output ---");

$display("String Value: %s", my_string);

$display("Data_Val = %h", Data_Val);

$display("data_val = %h", data_val);

$display("Mixed Logic = %b", mixed_logic);

$display("Analog Voltage = %f", analog_voltage);

$finish;

end

endmodule
```